

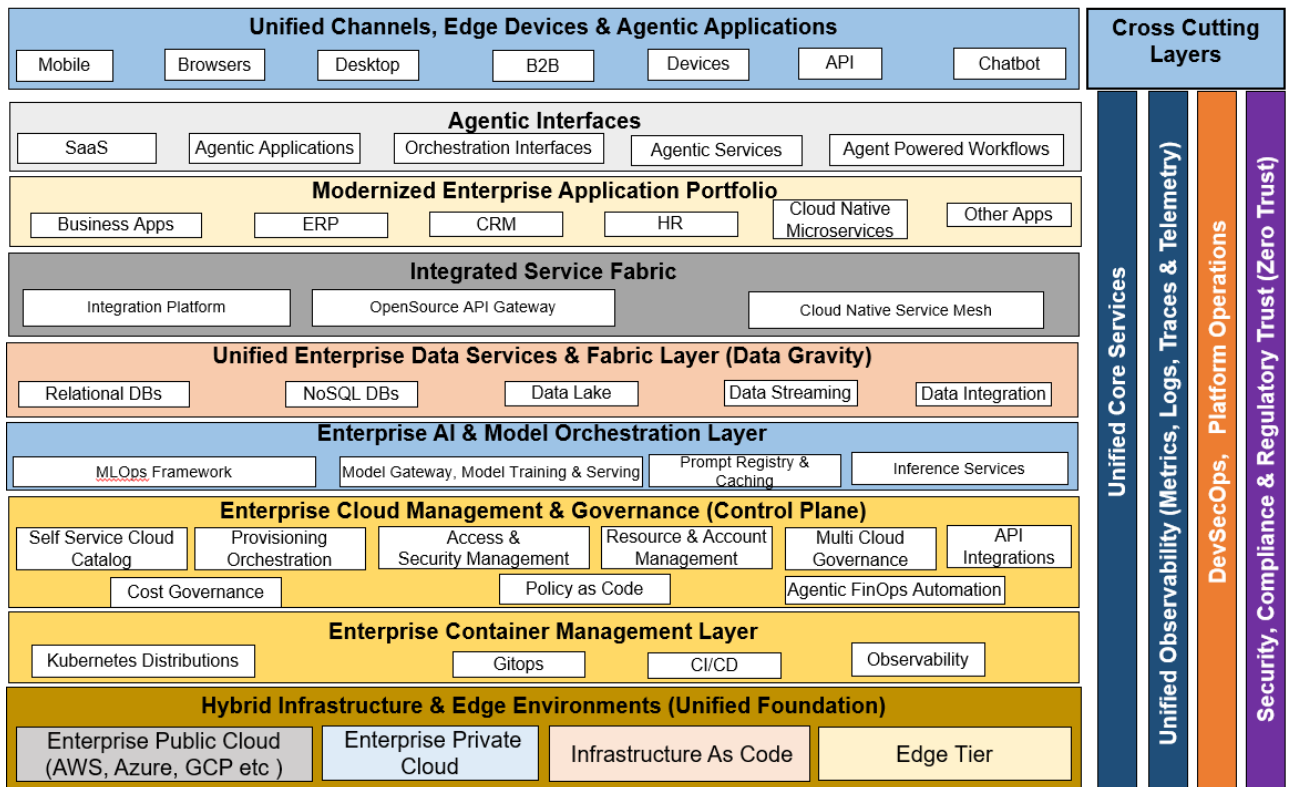


Figure 1: Hybrid cloud management model

observability within the container ecosystem.

- **Integration platform (e.g., Apache Camel):** Provides lightweight, enterprise integration patterns to connect diverse systems to fetch historical data.

Unified enterprise data services and fabric layer: This layer addresses data gravity and provides a standardised approach to data management across the hybrid landscape. It ensures that data can be discovered, accessed, and governed regardless of where it is stored. Various data services are:

- Relational database; e.g., PostgreSQL
- NoSQL database; e.g., Cassandra, Redis
- Data lake for storing and managing large volumes of unstructured data; e.g., MinIO, Apache Hive
- Data streaming for real-time, event-driven architecture; e.g., Apache Kafka

- Data integration for change data capture; e.g., Debezium

Enterprise AI and model

orchestration layer: This layer provides the standard tools needed to train, serve, and govern models. Key components include:

- **MLOps framework:** To orchestrate the end-to-end machine learning lifecycle; e.g., Kubeflow, MLflow.
- **Model gateway:** Intelligently routes requests to the most efficient model using local models for routine tasks and public LLMs for complex reasoning.
- **Model training and serving:** Develops and exposes models as scalable services; e.g., TensorFlow, Seldon Core.
- **Prompt registry and caching:** Caches AI responses, significantly reducing API costs and latency; e.g., Valkey.
- **Inference services:** Runs AI models on private hardware for maximum data security; e.g., vLLM, KServe.

Enterprise cloud management

and governance: This enables the provisioning of infrastructure components, platform services components, security and access management, and multi-cloud governance. This platform provides a self-service catalogue for users that helps them manage their services. It manages costs (FinOps) and handles Policy-as-Code (ensuring no one creates an insecure public database).

This is the single-pane-of-glass that orchestrates operations, controls costs, and enforces policies across the entire hybrid cloud.

- **Policy-as-Code:** Provides a unified way to enforce security policies and regulatory compliance; e.g., Open Policy Agent (OPA), Kyverno.
- **Cost governance:** This addresses cloud cost optimisation automation, and provides real-time visibility into Kubernetes spending; e.g., OpenCost.